3 Real numbers and intervals

Sheet 13 Thu 19 Nov. 2026 *Sets of numbers:* $\mathbb{N}, \mathbb{Z}, \mathbb{D}, \mathbb{Q}, \mathbb{R}$

1. Warm-up · Quick [NUM] Give the decimal form of $\frac{3}{8}$ and an approximation of $\frac{5}{6}$ to within 10^{-3}.
   Ans. 0.375; 0.833.
2. Core · Method [NUM] Give the smallest set $(\mathbb{N}, \mathbb{Z}, \mathbb{D}, \mathbb{Q}, \mathbb{R})$ containing each of these numbers: -4; 0.25; $\frac{1}{3}$; $\sqrt{2}$; $\frac{6}{3}$.
   Ans. $-4 \in \mathbb{Z}$; $0.25 \in \mathbb{D}$; $\frac{1}{3} \in \mathbb{Q}$; $\sqrt{2} \in \mathbb{R}$; $\frac{6}{3} = 2 \in \mathbb{N}$.
   The smallest set is the key: $\frac{6}{3}$ *is a natural number.*
3. Core + · True/false [NUM] True or false? Justify. The number $\frac{1}{3}$ is a decimal number.
   Ans. False. If there were integers a and n with $\frac{1}{3} = \frac{a}{10^n}$, then $10^n = 3a$ would be divisible by 3: impossible (the sum of the digits of 10^n is 1).
4. Review · Quick [SUB] Calculate $(x+1)(x-3)$ for $x = 2$.
   Ans. -3.
5. Thread · Quick [SOL] How many faces, vertices and edges does a right triangular prism have?
   Ans. 5 faces (2 triangles, 3 rectangles), 6 vertices, 9 edges.

Sheet 14 Mon 23 Nov. 2026 *Decimal bounds and rounding*

1. Warm-up · Quick [NUM] Round 3.14159 to the nearest 10^{-2} and then to the nearest 10^{-3}.
   Ans. 3.14 then 3.142.
   The next digit decides: $1 < 5$ *then* $5 \geqslant 5$.
2. Core · Method [NUM] Bound $\sqrt{2}$ between two decimals that differ by 10^{-2}.
   Ans. $1.41^2 = 1.9881 < 2 < 2.0164 = 1.42^2$: $1.41 < \sqrt{2} < 1.42$.
3. Core + · Code reading [ALG] What is the final value of `k`? What can be deduced about $\sqrt{2}$?

```
k = 10
while (k/10)**2 < 2:
    k = k + 1
```

   Ans. $k = 15$: $1.4^2 = 1.96 < 2$ and $1.5^2 = 2.25 \geqslant 2$, so $1.4 < \sqrt{2} \leqslant 1.5$.
4. Review · Quick [QP] 5 kg of rice cost 12,500 Ar. How much do 8 kg cost?
   Ans. $\frac{12{,}500}{5} \times 8 = 20{,}000$ Ar.
5. Thread · True/false [DEM] True or false? Justify. A quadrilateral whose diagonals have equal length is a rectangle.
   Ans. False. Counter-example: an isosceles trapezium has diagonals of equal length without being a rectangle.

Sheet 15 Tue 24 Nov. 2026 *Irrational numbers:* $\frac{1}{3}$ *is not decimal,* $\sqrt{2}$ *is irrational*

1. Warm-up · Quick [CAL] Write $\sqrt{12}$ and $\sqrt{50}$ in the form $a\sqrt{b}$ with b the smallest possible integer.
   Ans. $\sqrt{12} = 2\sqrt{3}$; $\sqrt{50} = 5\sqrt{2}$.
2. Core · Sorting [NUM] Among $0.333\ldots$; $\sqrt{9}$; $\sqrt{2}$; π; $\frac{22}{7}$, which are rational?
   Ans. $0.333\ldots = \frac{1}{3}$; $\sqrt{9} = 3$; $\frac{22}{7}$: rational. $\sqrt{2}$ and π are irrational.

$\frac{22}{7}$ *is only an approximation of* π*, not equal to* π*.*

3. Core + · Reasoning [DEM] Suppose $\sqrt{2} = \frac{p}{q}$ with $\frac{p}{q}$ in lowest terms, so $p^2 = 2q^2$. What can be said about the parity of p, then of q? What contradiction arises?

Ans. p^2 even, so p even, $p = 2k$; then $4k^2 = 2q^2$, $q^2 = 2k^2$: q even. Contradiction with the fraction being in lowest terms.

4. Review · Table [STA] Values 1 ; 2 ; 3 ; 4 with frequencies 5 ; 8 ; 4 ; 3. Give the total frequency, the relative frequency of the value 2 and the mean.

Value	1	2	3	4
Frequency	5	8	4	3

Ans. Total 20; relative frequency $\frac{8}{20} = 40\%$; mean $\frac{5+16+12+12}{20} = 2.25$.

5. Thread · Quick [REP] Give the coordinates of the reflection of $A(3,-2)$ in the y-axis.

Ans. $(-3,-2)$.

Pitfall: changing the sign of the ordinate (reflection in the x-axis).

Sheet 16 Thu 26 Nov. 2026 *Intervals of* $\mathbb{R}$*: notation and representation*

1. Warm-up · Quick [CMP] Compare -3 and -5, then $-\frac{1}{2}$ and -0.6.

Ans. $-3 > -5$; $-0.5 > -0.6$.

For negative numbers, the one with the smaller absolute value is the greater.

2. Core · Method [NUM] Write as an interval: $-2 \leqslant x < 5$; $x > 3$; $x \leqslant 0$.

Ans. $[-2,5)$; $(3,+\infty)$; $(-\infty,0]$.

The bracket faces inwards if the endpoint is included; ∞ is always excluded.

3. Core + · MCQ [NUM] The intersection $(-\infty,2) \cap [-1,4]$ is:

A. $(2,4]$ **B.** $(-\infty,4]$ **C.** $[-1,2)$ **D.** $[-1,2]$

Ans. C. $[-1,2)$

Intersection: values satisfying both conditions $x < 2$ and $-1 \leqslant x \leqslant 4$.

4. Review · Table [LEC] Journey times (min).

Class	$[0,10)$	$[10,20)$	$[20,30)$
Frequency	4	9	7

What are the total frequency, the modal class and the relative frequency of $[10,20)$?

Ans. 20; $[10,20)$; $\frac{9}{20} = 45\%$.

5. Thread · Problem [DIV] One bus runs every 15 min, another every 20 min. They are together at 8 a.m. At what time are they next together?

Ans. $\text{lcm}(15,20) = 60$: at 9 a.m.

Sheet 17 Mon 30 Nov. 2026 *Absolute value and distance between two real numbers*

1. Warm-up · Quick [REP] What is the distance between -3 and 5 on the number line?

Ans. $|5-(-3)| = 8$.

2. Core · Quick [CAL] Calculate $|-7|$, $|3-8|$ and $|\pi-4|$.

Ans. 7; 5; $|\pi-4| = 4-\pi$ since $\pi < 4$.

3. Core + · Method [EQ] Solve $|x-3| = 2$.

Ans. The real numbers at distance 2 from 3: $x = 1$ or $x = 5$.

Geometric interpretation rather than case by case.

4. Review · Formula [IMG] $f(x) = x^2 - 3$. Give the image of -2 and the preimages of 6.
   Ans. $f(-2) = 1$; $x^2 - 3 = 6 \Leftrightarrow x^2 = 9$: preimages -3 and 3.
5. Thread · Formula → table [REG] $f(x) = -x + 3$. Complete: $f(-1)$, $f(0)$, $f(2)$.
   Ans. 4 ; 3 ; 1.

Sheet 18 Tue 1st Dec. 2026 *Inequality $|x - a| \leqslant r$ and the associated interval*

1. Warm-up · Quick [EQ] Solve $2x - 3 \leqslant 5$.
   Ans. $2x \leqslant 8$, $x \leqslant 4$: $S = (-\infty, 4]$.
2. Core · Method [EQ] Solve $|x - 2| \leqslant 3$ and give S as an interval.
   Ans. The real numbers at distance at most 3 from 2: $S = [-1, 5]$.
   Check: $|-1 - 2| = 3$ *and* $|5 - 2| = 3$.
3. Core + · Translate [NUM] Express as an inequality with absolute value: $x \in [-1, 5]$ then $x \in (4, 10)$.
   Ans. $|x - 2| \leqslant 3$ (centre 2, radius 3); $|x - 7| < 3$ (centre 7, radius 3).
4. Review · Problem [CMP] Pack A: 6 yoghurts for €4.20. Pack B: 10 yoghurts for €6.80. Which is better value?
   Ans. A: €0.70 each; B: €0.68: B.
5. Thread · Quick [GEO] A disc has radius 5 cm. Give its area and its circumference (exact values).
   Ans. Area $= \pi \times 5^2 = 25\pi$ cm^2; circumference $= 2\pi \times 5 = 10\pi$ cm.